

EvoLab Project Guide

Biology primer, simulation pipeline, system architecture, data flow, and visualization guide

Purpose: an interactive teaching model that evolves a synthetic diploid population, records parentage, summarizes population genetics, and infers a UPGMA tree from genotype distances.

What this guide covers

Part	What you will learn
Biology foundations	DNA, genes, alleles, diploidy, genotype, phenotype, generations, and inheritance.
Project pipeline	How one cohort becomes the next and where mutation, selection, and ancestry are recorded.
System design	Responsibilities of the Java packages and how snapshots move to analysis and UI.
Species and generations	How to use the kingdom, subcategory, species, and generation controls.
Reading the output	What UPGMA, genotype distribution, allele frequency, and heterozygosity mean.
Class diagram	Actual Java inheritance and interface implementations, plus important data relationships.

Model boundary

Animal and plant entries are selectable species labels for synthetic experiments. The application does not load real genomes for those species. All profiles currently share one diploid trait locus, the same inheritance engine, and the same simplified selection rule. The labels let a learner organize and compare runs; they do not represent empirical species-specific biology.

Project root: Digital-Evolution-Phylogenetics-Laboratory | Guide prepared 5 October 2026

1. Biology foundations

The simulation is easier to follow if we separate the information stored in DNA from the visible trait and from the population-level statistics.

Term	Plain-language meaning in EvoLab
DNA and gene	DNA carries inherited information. A gene is a modeled locus associated here with one trait.
Allele	A variant at that locus. EvoLab currently labels the two variants Light and Dark and assigns values 0 and 1.
Diploid	An organism has two allele copies at each modeled locus, one inherited from each parent in this sexual-reproduction abstraction.
Genotype	The pair of alleles carried by one organism, such as Dark/Light.
Phenotype	The expressed trait value computed from the genotype. The current additive rule averages the two allele values.
Generation	One discrete cohort. Parents from one cohort create a same-sized offspring cohort for the next generation.
Population	The group of organisms evolving together under the same simple environment and selection rule.
Pedigree	A record of who the two parents of each simulated offspring were. It is a network because each child has two parents.

How inheritance works here

For each locus, each parent randomly transmits one of its two alleles. Those two transmitted alleles form the offspring genotype. Each transmitted allele can mutate independently. This is a useful simplified model of segregation, but it does not implement real chromosome counts, linkage, recombination/crossover, sex-specific biology, or a plant life cycle.

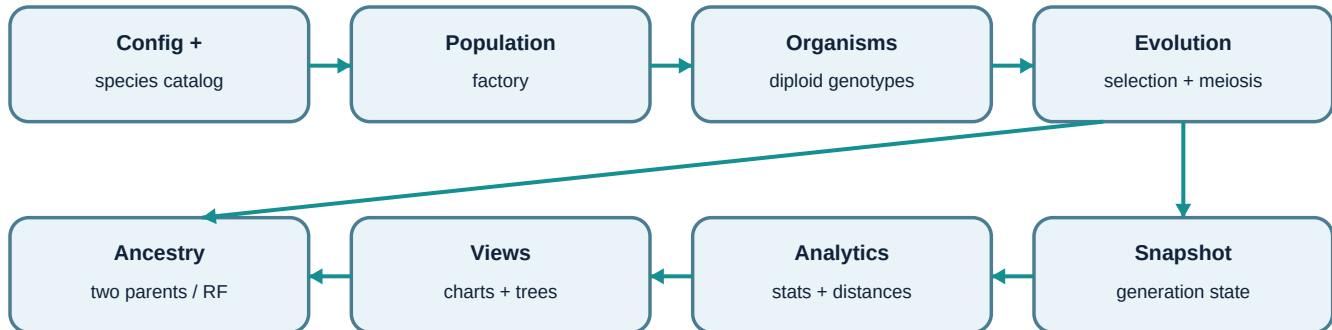
Use case

EvoLab is intended for learning and experimentation: observe how inheritance, mutation, and selection can change trait frequencies over generations; compare population summaries; inspect a recorded family line; and compare a distance-based inferred tree with one declared ancestry projection. It is not a species identification or conservation analysis tool.

Key distinction: Animal and Plant are profile choices, not different biological simulators. In this version, both use diploid inheritance and the same engine.

2. Simulation pipeline

A run starts from configuration and a selected synthetic species profile. One engine step replaces the whole cohort with a new cohort of the same size.



Step	Operation	Output
1. Configure	Read population size, starting Dark-allele frequency, mutation rate, seed, and terrain. Load species categories and trait-locus metadata from the CSV catalog.	Repeatable settings + selected profile.
2. Founders	PopulationFactory creates two alleles at the chosen profile's one trait locus for each founder, using the configured seed.	Generation 0 cohort.
3. Score and select	FitnessFunction compares the expressed trait value with the terrain target. EvolutionEngine samples two distinct reproductively mature parents, weighted by fitness.	Selected parent pairs.
4. Inherit and mutate	MeiosisReproduction chooses one allele per parent at each locus and applies independent mutation probability.	Diploid offspring genotypes.
5. Record and publish	AncestryRecorder stores both parent IDs. The engine replaces the cohort and publishes a SimulationSnapshot.	Generation history and observer event.
6. Analyze and render	The tracker calculates population genetics; DistanceMatrix and UPGMA infer a tree; JavaFX updates the overview and comparison views.	Charts, ecosystem, tree, and pedigree.

Determinism

A fixed random seed makes the same profile/configuration reproducible in this implementation. Changing the selected species resets to generation 0. Historical snapshots are retained in memory for the currently active profile during the app session; they are not persisted after the application closes.

3. System design and data flow

The code separates the simulation model from JavaFX rendering. The engine publishes a snapshot; analysis modules and the UI consume it without the engine drawing charts itself.

Layer / package	Key classes	Responsibility
Configuration	config.SimulationConfig	Validate simulation parameters.
Profiles	core.SpeciesCatalog, SpeciesProfile, SpeciesType	Load Animal/Plant labels, subcategories, scientific names, and synthetic trait-locus names.
Core + genetics	Organism, Genotype, Gene, Allele, ExpressionRule	Represent organisms and inherited trait information.
Evolution	PopulationFactory, EvolutionEngine, FitnessFunction, ExperimentController	Create founders, choose parents, reproduce, score fitness, and control runs.
Ancestry	AncestryRecorder.BirthRecord	Keep both parent links; provide a parent-1 projection for RF tree comparison.
Population analytics	PopulationGeneticsTracker	Track per-generation allele/genotype statistics and Hardy-Weinberg expectations.
Phylogenetics	DistanceMatrix, UPGMAAlgorithm, PhylogeneticTree, TreeComparator	Compute genome distances, infer a tree, serialize Newick, and calculate RF distance.
Presentation	MainApp, EcosystemPane, TreePane, StatsPane, GenerationComparisonPane	Show the live overview, historical comparison, and on-demand two-parent family hierarchy.

Data flow

The configuration and selected catalog row feed PopulationFactory. Organisms carry a SpeciesProfile and Genotype. EvolutionEngine reads the current population, scores and selects parents, calls the reproduction strategy, records each birth in AncestryRecorder, and emits SimulationSnapshot. MainApp stores that snapshot by generation and sends it to the ecosystem, statistics, and tree views. PopulationGeneticsTracker derives frequencies and genotype proportions. DistanceMatrix turns organism genotypes into pairwise distances; UPGMAAlgorithm turns those distances into a PhylogeneticTree. GenerationComparisonPane compares one saved snapshot with the live one and can expand the selected organisms' parent network when requested.

The UPGMA reference tree is inferred from genome distances. The pedigree is separately recorded. They are different objects and must not be treated as the same tree.

4. Species catalog and generation checks

The header filters a catalog in three stages: Kingdom (Animal or Plant), Subcategory, then Species. The catalog currently includes 18 selectable names across mammals, birds, reptiles, insects, amphibians, fish, flowering plants, conifers, ferns, and mosses.

Kingdom	Subcategory	Common name	Scientific name	Synthetic locus
Animal	Mammals	Gray wolf	Canis lupus	coat
Animal	Mammals	Red fox	Vulpes vulpes	coat
Animal	Mammals	Snowshoe hare	Lepus americanus	coat
Animal	Mammals	African savanna elephant	Loxodonta africana	coat
Animal	Birds	Rock pigeon	Columba livia	plumage
Animal	Birds	Emperor penguin	Aptenodytes forsteri	plumage
Animal	Reptiles	Green sea turtle	Chelonia mydas	scalePigment
Animal	Insects	Western honey bee	Apis mellifera	bodyColor
Animal	Amphibians	Axolotl	Ambystoma mexicanum	skinPigment
Animal	Fish	Zebrafish	Danio rerio	scalePigment
Plant	Flowering plants	Thale cress	Arabidopsis thaliana	leafPigment
Plant	Flowering plants	Maize	Zea mays	leafPigment
Plant	Flowering plants	Sunflower	Helianthus annuus	leafPigment
Plant	Flowering plants	Asian rice	Oryza sativa	leafPigment
Plant	Flowering plants	Pedunculate oak	Quercus robur	leafPigment
Plant	Conifers	Scots pine	Pinus sylvestris	needlePigment
Plant	Ferns	Bracken fern	Pteridium aquilinum	frondPigment
Plant	Mosses	Ceratodon moss	Ceratodon purpureus	leafPigment

Workflow

1. Choose an Animal or Plant kingdom, then a subcategory and species. This starts a fresh synthetic population for that profile. 2. Run generations with Run 1, Run 10, Run N, or Start auto. 3. The header shows the live generation. Select any saved generation and press Inspect to view that snapshot; Return to live restores the current one. 4. Open Generation comparison, choose an earlier generation and one organism from each cohort, then compare population summaries and genotypes. Use Build full family hierarchies to display both recorded parent networks.

Cross-check means comparing two generations within a selected synthetic species profile. This version does not crossbreed different catalog species or compare empirical sequence datasets.

Taxonomic spellings are catalog labels; the genomes and trait values are generated by the simulator. Example names are cross-checkable in the GBIF Backbone Taxonomy [1-4].

5. How to read the visualizations

UPGMA inferred tree

UPGMA means Unweighted Pair Group Method with Arithmetic Mean. It starts with each organism as a separate cluster, merges the two clusters with the smallest average pairwise distance, updates distances by the arithmetic mean weighted by cluster membership, and repeats until one tree remains. In EvoLab, the distance matrix compares unordered diploid allele pairs. The radial drawing is a genome-distance reconstruction, not the recorded family tree. Classical UPGMA assumes an approximately clock-like / ultrametric distance pattern; selection, mutation, drift, and one-locus data can violate that assumption. RF compares the inferred tree to a parent-1-only projection, so it omits each organism's second parent.

Genotype distribution

The bar chart counts how many organisms carry each observed allele pair at the trait locus: for example, Light/Light, Light/Dark, and Dark/Dark. Counts are the observed genotype proportions multiplied by population size. It shows the make-up of the current selected generation, not an estimate of all real individuals of that species.

Allele frequency and heterozygosity

Let p be the frequency of the Dark allele and $q = 1 - p$ be the Light-allele frequency. Under Hardy-Weinberg equilibrium for a two-allele diploid locus, expected genotype proportions are p^2 (Dark/Dark), $2pq$ (Dark/Light), and q^2 (Light/Light). Expected heterozygosity is $H_E = 2pq$. Observed heterozygosity H_O is the fraction of sampled organisms whose two alleles differ. The line chart plots Dark-allele frequency, H_E , and H_O by generation. A difference $H_O - H_E$ is a descriptive deviation; it is not by itself proof of a cause.

Example	Calculation	Meaning
$p = 0.6, q = 0.4$	Expected heterozygosity = $2 \times 0.6 \times 0.4 = 0.48$	About 48% heterozygotes expected under the model assumptions.
Observed sample	If 12 of 30 organisms are heterozygous, $H_O = 12/30 = 0.40$	Observed is 8 percentage points below expectation in this sample.

A small finite population, selection, mutation, and the simulation's particular mating process can move H_O away from H_E . The chart is an educational diagnostic, not a formal population-genetic study.

6. Class hierarchy image

The image distinguishes Java inheritance (extends), interface implementations (implements), and selected teal data relationships. Most domain classes use composition rather than subclassing. Animal and Plant are enum values with profile records; they are not subclasses of Organism.

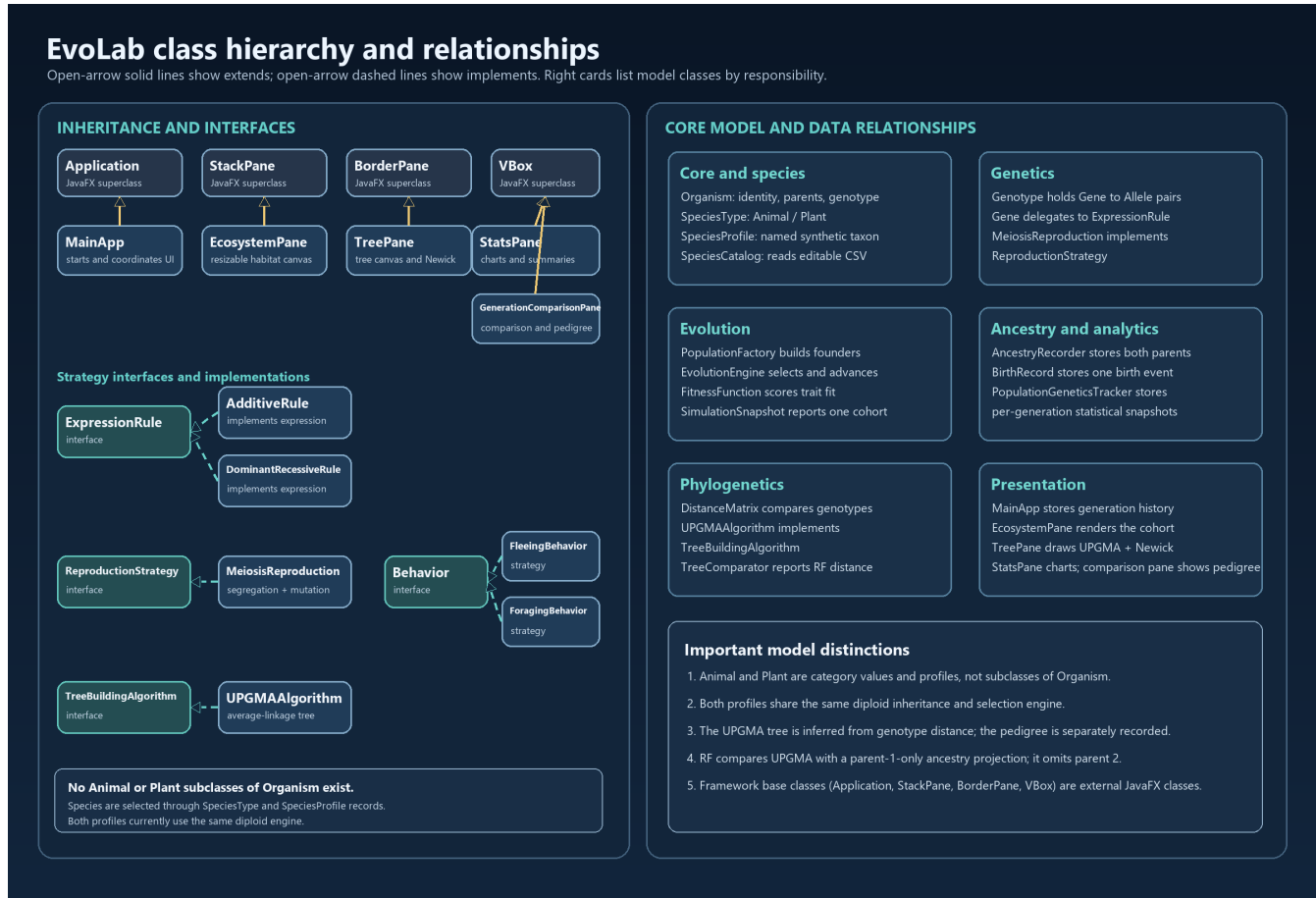


Diagram source: current Java declarations in src/core, src/genetics, src/evolution, src/phylogenetics, and src/visualization. JavaFX superclass boxes are framework types. Nested BirthRecord, GenerationSnapshot, and TreeNode are represented within their owning model classes in the project code.

7. Scope, limitations, and sources

Current scientific/model limitations

- One synthetic diploid trait locus per selected profile; no real sequence data. - All catalog species share the same reproduction and life-stage rules; the model does not implement haploidy, alternation of generations, pollination, self-fertilization, or species-specific life histories. - The catalog supports selection and labeling only; different species cannot interbreed in this version. - The current cohort history is stored in memory for the active profile and disappears when the app closes or the selected profile changes. - The full two-parent family network is available on demand; RF uses only a parent-1 projection. - UPGMA results from one low-resolution locus should be used for demonstration, not empirical phylogenetic claims.

Primary taxonomic references

[1] GBIF Backbone Taxonomy, *Canis lupus*: <https://www.gbif.org/taxon/QLXL>

[2] GBIF / Catalogue of Life, *Vulpes vulpes*: <https://www.gbif.org/taxon/5BSG3>

[3] GBIF Backbone Taxonomy, *Arabidopsis thaliana*: <https://www.gbif.org/species/3052436/metrics>

[4] GBIF Backbone Taxonomy, *Zea mays* and *Quercus robur*: <https://www.gbif.org/taxon/5CXX5> and <https://www.gbif.org/taxon/4R5YN>

For a research-grade extension, replace synthetic profiles with a sourced dataset that defines accession/sample IDs, sequence or genotype provenance, ploidy, locus definitions, collection metadata, and licensing. Then make the model's inheritance rules match the organism being represented.

Project files

config/simulation.properties - simulation defaults. config/species-catalog.csv - editable species labels, subcategories, scientific names, and synthetic trait-locus names. docs/architecture.md - architecture and data flow. docs/project-breakdown.md - package and class responsibilities.